Electrical & Computer Engineering Department

Ms. Computer Vision

University of Central Florida

Orlando, Florida

Medical Image Computing

Programming Assignment #1

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2026 – 2027

https://github.com/DarinkaTownsend/Deep-Learning-based-Pneumonia-Detection-Using-Chest-X-Ray-Images.git

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1. **Objetive** 3
2. **Introduction** 3
3. **Data Set** 3
4. **Task 1.1 Scratch** 5
5. **Task 1.2 – Balanced Training** 6
6. **Task 1.2.1 – Balanced Training with WeightedRandomSampler (Extra Experiment)** 7
7. **Comparision** 8
8. **Github Repository** 10
9. **Conclusion** 10
10. **References** 11

1. Objetive

The objective of this assignment is to develop and evaluate a deep learning model for binary pneumonia classification using pediatric chest X-ray images. Specifically, it compares training a ResNet-18 model from scratch versus fine-tuning a pretrained ImageNet model to analyze the impact of transfer learning on performance and generalization

2. Introduction

Pneumonia remains one of the leading causes of mortality among young children worldwide, highlighting the need for timely and accurate diagnosis (Rudan et al., 2008; World Health Organization [WHO], 2019). Chest X-ray imaging is widely used in clinical practice to detect pulmonary abnormalities; however, manual interpretation can be challenging due to subtle visual patterns and inter-observer variability, particularly in pediatric cases (Rajpurkar et al., 2017). In recent years, automated image analysis systems based on deep learning have demonstrated strong potential in supporting radiological decision-making and improving diagnostic consistency (Esteva et al., 2019; Rajpurkar et al., 2017).

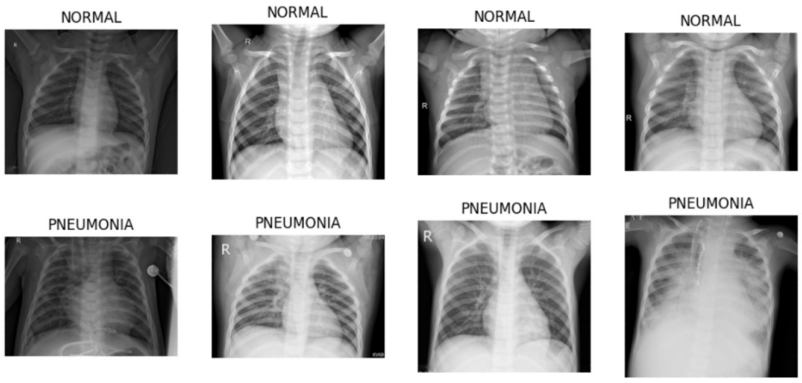
In this study, we investigate the use of a convolutional neural network (ResNet-18) for binary classification of pediatric chest X-ray images (Normal vs. Pneumonia). Two training strategies are evaluated: training from scratch with random initialization and fine-tuning a model pretrained on ImageNet. Transfer learning has been shown to improve convergence stability and generalization in medical imaging tasks with limited labeled data (Shin et al., 2016), and this work analyzes its impact on classification accuracy and model robustness in a pediatric pneumonia detection setting.

3. Data Set

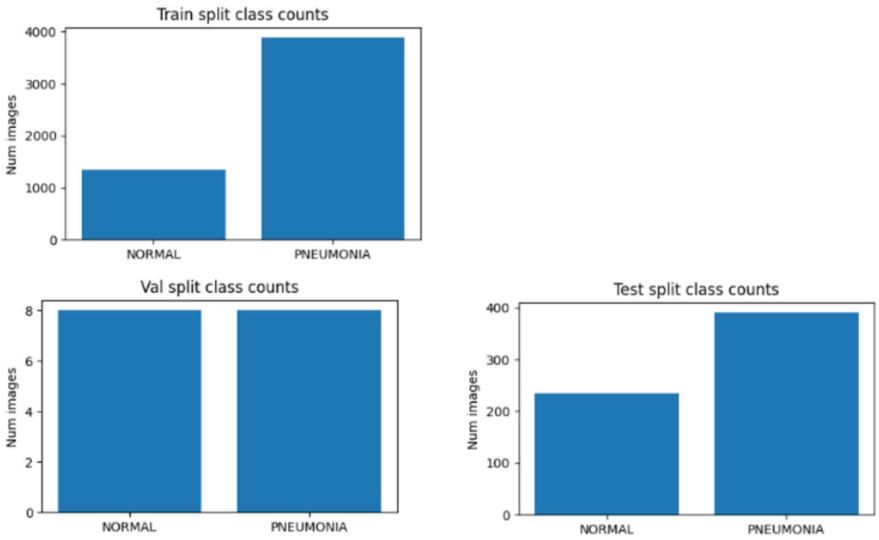
Pneumonia remains one of the leading causes of mortality among young children worldwide, highlighting the need for timely and accurate diagnosis. Chest X-ray imaging is widely used in clinical practice to detect pulmonary abnormalities; however, manual interpretation can be challenging due to subtle visual patterns and inter-observer variability. Automated image analysis systems based on deep learning have emerged as promising tools to support radiological decision-making.

In this study, we investigate the use of a convolutional neural network (ResNet-18) for binary classification of pediatric chest X-ray images (Normal vs. Pneumonia). Two training strategies are evaluated: training from scratch with random initialization and fine-tuning a

model pretrained on ImageNet. The objective is to analyze the impact of transfer learning on convergence behavior, classification accuracy, and generalization performance in a medical imaging context.



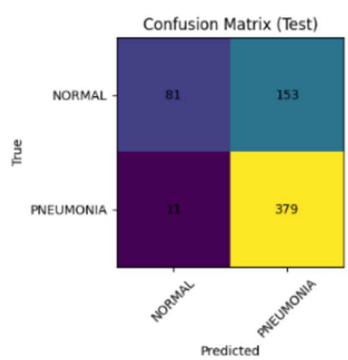
The dataset presents a noticeable class imbalance, particularly in the training and test splits. In the training set, the number of Pneumonia images is significantly higher than Normal cases (approximately 3,800 vs. 1,300), indicating that the model is exposed to substantially more positive samples during learning. The validation set is balanced, containing an equal number of Normal and Pneumonia images (8 per class), while the test set also shows imbalance, with more Pneumonia cases than Normal. This distribution may bias the model toward predicting the majority class and should be considered when interpreting performance metrics.



4. Task 1.1 Scratch

Training the ResNet-18 model from scratch reveals a clear imbalance between training behavior and generalization performance. The training loss decreases rapidly and stabilizes at a very low value, while training accuracy remains consistently high (around 95–96%), indicating that the model is able to fit the training data effectively. However, the validation curves tell a different story. The validation loss fluctuates significantly across epochs and does not show a stable downward trend. Similarly, validation accuracy is unstable and remains considerably lower than training accuracy. This divergence between training and validation performance suggests strong overfitting.

The confusion matrix further clarifies the model's behavior. While the model correctly identifies most Pneumonia cases (379 true positives), it struggles significantly with Normal cases, misclassifying a large portion of them (153 false positives). This imbalance is reflected in the class-wise metrics: Pneumonia achieves a high recall (0.9718), meaning the model rarely misses pneumonia cases, but the recall for Normal is very low (0.3462). In practice, this means the model has learned to heavily favor predicting the majority class (Pneumonia), likely influenced by the training data imbalance. The relatively high precision for Normal (0.8804) indicates that when the model predicts Normal, it is often correct, but it rarely makes that prediction.




```
Class-wise Metrics:
NORMAL:
  Precision: 0.8804
  Recall:    0.3462
  F1-score:  0.4969
PNEUMONIA:
  Precision: 0.7124
  Recall:    0.9718
  F1-score:  0.8221

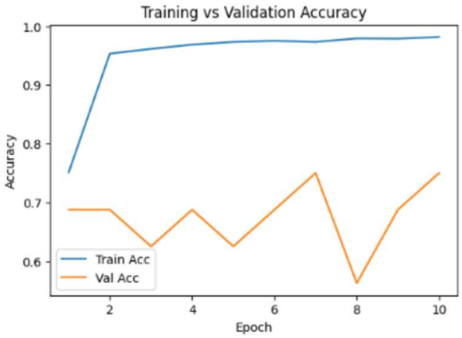
Macro Average:
  Precision: 0.7964
  Recall:    0.6590
  F1-score:  0.6595
```

The macro-averaged F1-score (0.6595) confirms that overall performance is moderate and skewed. Although the overall accuracy may appear acceptable due to the dominance of Pneumonia samples, the poor recall for Normal demonstrates limited class balance and weak generalization. This pattern is typical when training from scratch on a relatively small and imbalanced medical dataset: the network does not learn sufficiently robust feature representations and instead adapts strongly to the dominant class distribution.

In summary, Task 1.1 shows that training from random initialization leads to overfitting and class bias. While the model captures discriminative patterns for Pneumonia, it fails to generalize well to minority-class samples. These results highlight the importance of transfer learning and regularization strategies when working with limited medical imaging datasets.

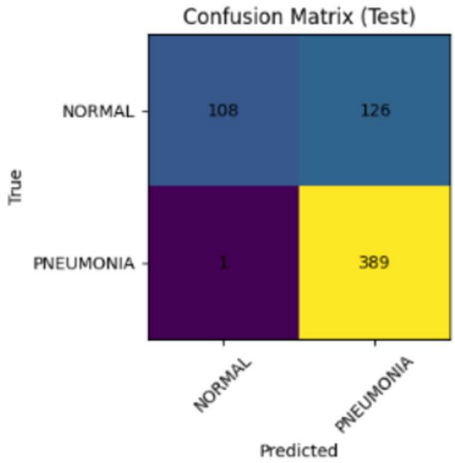
5. Task 1.2 – Balanced Training

When fine-tuning the pretrained ResNet-18, the training dynamics change noticeably compared to training from scratch. The training loss decreases rapidly and stabilizes at a very low value, while training accuracy approaches 98%, indicating strong fitting capacity. More importantly, although the validation curves still fluctuate, they show improved stability and higher overall accuracy compared to Task 1.1. The validation accuracy oscillates between roughly 63% and 75%, which, despite variability due to the very small validation set, remains consistently higher than the scratch configuration. This suggests that pretrained weights provide a better initialization, allowing the model to converge toward more discriminative representations.



The confusion matrix highlights a substantial improvement in class balance. The model correctly classifies 389 Pneumonia cases and only misclassifies 1 as Normal, yielding an almost perfect recall for Pneumonia (0.9974). While the model still struggles with Normal cases (recall = 0.4615), it performs better than in Task 1.1, correctly identifying 108 Normal images. The macro F1-score increases to 0.7447, reflecting improved overall balance

between classes. Precision for Normal is extremely high (0.9908), indicating that when the model predicts Normal, it is almost always correct; however, the relatively low recall suggests it remains conservative in assigning that label.



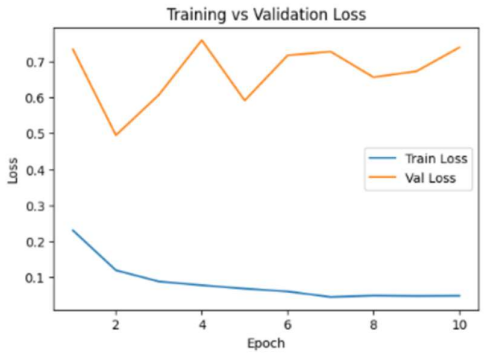

```
Class-wise Metrics:
NORMAL:
  Precision: 0.9908
  Recall:    0.4615
  F1-score:  0.6297
PNEUMONIA:
  Precision: 0.7553
  Recall:    0.9974
  F1-score:  0.8597

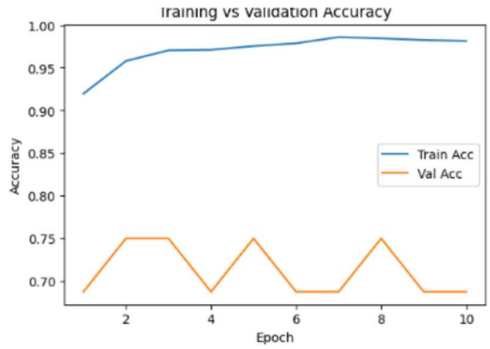
Macro Average:
  Precision: 0.8731
  Recall:    0.7295
  F1-score:  0.7447
```

Overall, transfer learning significantly enhances performance by improving feature generalization and reducing class bias. The pretrained model captures more robust low-level and mid-level representations learned from large-scale natural images, which appear transferable to medical imaging despite domain differences. While class imbalance still influences predictions, the pretrained configuration demonstrates better generalization and a stronger macro-level performance, confirming the benefit of transfer learning in small, imbalanced medical datasets.

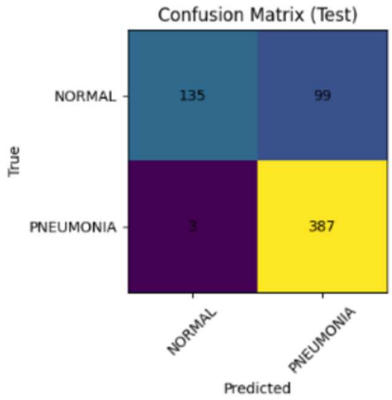
6. Task 1.2.1 – Balanced Training with WeightedRandomSampler (Extra Experiment)

To further address the class imbalance observed in previous experiments, a WeightedRandomSampler was incorporated during training to enforce a more balanced class distribution per batch. Unlike simple fine-tuning, this strategy directly modifies the sampling process, ensuring that minority class samples (Normal) are seen more frequently during optimization. The training curves show stable convergence, with low training loss and high training accuracy (~98%), similar to standard transfer learning. Validation accuracy remains more consistent compared to Task 1.1, fluctuating around 69–75%, but without the severe instability observed in the scratch model.





The confusion matrix demonstrates a meaningful shift in model behavior. The number of correctly classified Normal cases increases to 135 (compared to 108 in standard fine-tuning), significantly improving Normal recall to 0.5769. At the same time, Pneumonia recall remains extremely high (0.9923), with only 3 false negatives. This balance between classes is reflected in the macro F1-score of 0.8047, the highest among all experiments. Importantly, overall accuracy (83.65%) remains strong while reducing bias toward the majority class.




```
Class-wise Metrics:
NORMAL:
  Precision: 0.9783
  Recall:    0.5769
  F1-score:  0.7258
PNEUMONIA:
  Precision: 0.7963
  Recall:    0.9923
  F1-score:  0.8836

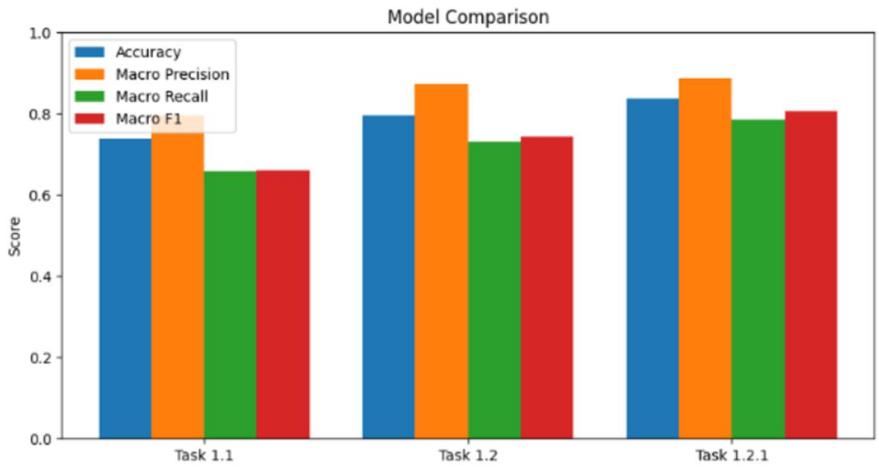
Macro Average:
  Precision: 0.8873
  Recall:    0.7846
  F1-score:  0.8047
```

These results indicate that balanced sampling effectively mitigates class bias without sacrificing sensitivity to Pneumonia cases. In medical diagnosis, maintaining high Pneumonia recall is critical to minimize false negatives, and this approach achieves that while substantially improving minority class detection. Among all tested configurations, this balanced transfer learning setup provides the most robust and clinically reliable performance.

7. Comparision

Across the three configurations, a clear performance progression is observed. Training from scratch (Task 1.1) yields the lowest overall performance, with 73.7% accuracy and a macro

F1-score of 0.6595, reflecting strong class bias and limited generalization. Introducing transfer learning (Task 1.2) significantly improves results, increasing accuracy to 79.7% and macro F1 to 0.7447, demonstrating the benefit of pretrained feature representations. The balanced transfer learning approach (Task 1.2.1) achieves the best overall performance, with 83.7% accuracy and a macro F1-score of 0.8047, indicating improved class balance and reduced bias. Notably, while Pneumonia recall remains consistently high across all setups, the balanced strategy substantially enhances Normal class detection without sacrificing sensitivity, making it the most robust and clinically reliable configuration among the three.

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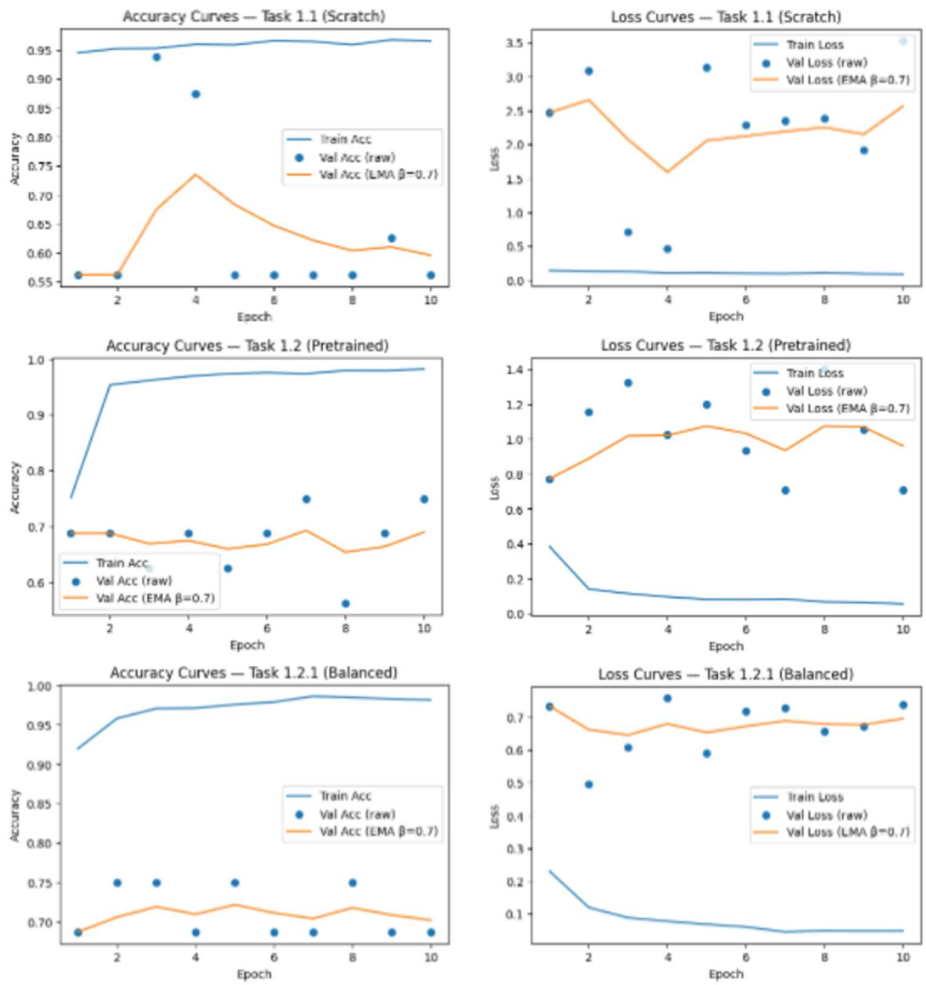
===== EXPERIMENT COMPARISON =====

Model          Acc       MacroPrec    MacroRec     MacroF1
-------------------------------------------------------------
Task 1.1       0.7372    0.7964       0.6590       0.6595
Task 1.2       0.7965    0.8731       0.7295       0.7447
Task 1.2.1     0.8365    0.8873       0.7846       0.8047
```



8. Github Repository

https://github.com/DarinkaTownsend/Deep-Learning-based-Pneumonia-Detection-Using-Chest-X-Ray-Images.git

9. Conclusion

In conclusion, this study demonstrates that transfer learning plays a critical role in improving pneumonia classification performance in limited and imbalanced medical imaging datasets. While training from scratch leads to overfitting and strong class bias, fine-tuning a pretrained ResNet-18 significantly enhances generalization and overall accuracy. Furthermore, incorporating balanced sampling through a WeightedRandomSampler provides the most robust results, improving minority class detection while maintaining very high Pneumonia recall. Overall, the balanced transfer learning approach offers the best

trade-off between sensitivity and class fairness, making it the most reliable configuration for this task.

10. References

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